

$$c) \quad z = 2x^2 + 3xy + 6x + 7y$$

$$\frac{\partial z}{\partial x} = 4x + 3y + 6 + 0$$

$$\frac{\partial z}{\partial y} = 0 + 3x + 0 + 7$$

$$= 3x + 7$$

$$\frac{\partial^2 z}{\partial x^2} = 4 + 0 + 0$$

$$= 4$$

$$\frac{\partial^2 z}{\partial y^2} = 0$$

$$d) \quad z = e^x \sin y + xy^2$$

$$\frac{\partial z}{\partial x} = e^x \sin y + y^2$$

$$f_{xy} = e^x \cos y + 2y$$

$$\frac{\partial z}{\partial y} = e^x \cos y + 2xy$$

$$f_{yx} = e^x \cos y + 2y$$

$$e) \quad z = x^2 y^3 + 3x^2 \sin y + e^y \cos^2 x$$

$$\frac{\partial z}{\partial x} = 2xy^3 + 6x \sin y - 2e^y \sin x$$

$$\frac{\partial^2 z}{\partial x^2} = 2y^3 + 6 \sin y - 2e^y \cos x$$

$$\frac{\partial z}{\partial y} = 3x^2 y^2 + 3x^2 \cos y + e^y \cos^2 x$$

$$\frac{\partial^2 z}{\partial y^2} = 6x^2 y + 3x^2 \sin y + e^y \cos^2 x$$

$$e) \quad z^2 - 3yz - 3x = 0$$

$$\text{prove } z_p = q$$

Implicit Differentiation

$$p = 3z^2 \frac{\partial z}{\partial x} - 3y \frac{\partial z}{\partial x} - 3 = 0$$

$$q = 3z^2 \frac{\partial z}{\partial y} - 3 \left(y \frac{\partial z}{\partial y} + z(1) \right) - 0 = 0$$

$$= 3z^2 \frac{\partial z}{\partial y} - 3y \frac{\partial z}{\partial y} + 3z = 0$$

$$3z^2 \frac{\partial z}{\partial x} - 3y \frac{\partial z}{\partial x} - 3 = 0$$

$$3z^2 \frac{\partial z}{\partial y} - 3y \frac{\partial z}{\partial y} + 3z = 0$$

$$3z^2 \frac{\partial z}{\partial x} - 3y \frac{\partial z}{\partial x} = 3$$

$$3z^2 \frac{\partial z}{\partial y} - 3y \frac{\partial z}{\partial y} = -3z$$

$$\frac{\partial z}{\partial x} (3z^2 - 3y) = 3$$

$$\frac{\partial z}{\partial y} (3z^2 - 3y) = -3z$$

$$\frac{\partial z}{\partial x} = \frac{3}{3z^2 - 3y}$$

$$q = \frac{\partial z}{\partial y} = \frac{-z}{z^2 - y}$$

$$p = \frac{\partial z}{\partial x} = \frac{1}{z^2 - y}$$

$$z \left(\frac{1}{z^2 - y} \right) = \frac{-z}{z^2 - y}$$